

SECRET

🔖 Tags:

Time Limit: 1 s Memory Limit: 128 MB
Submission: 13 AC: 6 Score: 99.77

Submit

Codes

AI Code Tips

Description

Dr.Kong is constructing a new machine and wishes to keep it secret as long as possible. He has hidden in it deep within some forest and needs to be able to get to the machine without being detected. He must make a total of T ($1 \leq T \leq 200$) trips to the machine during its construction. He has a secret tunnel that he uses only for the return trips.

The forest comprises N ($2 \leq N \leq 200$) landmarks (numbered $1..N$) connected by P ($1 \leq P \leq 40,000$) bidirectional trails (numbered $1..P$) and with a positive length that does not exceed 1,000,000. Multiple trails might join a pair of landmarks.

To minimize his chances of detection, Dr.Kong knows he cannot use any trail on the forest more than once and that he should try to use the shortest trails.

Help Dr.Kong get from the entrance (landmark 1) to the secret machine (landmark N) a total of T times. Find the minimum possible length of the longest single trail that he will have to use, subject to the constraint that he use no trail more than once.

(Note well: The goal is to minimize the length of the longest trail, not the sum of the trail lengths.)

It is guaranteed that Dr.Kong can make all T trips without reusing a trail.

Input

Line 1: Three space-separated integers: N , P , and T

Lines 2.. $P+1$: Line $i+1$ contains three space-separated integers, A_i , B_i , and L_i , indicating that a trail connects landmark A_i to landmark B_i with length L_i .

Output

Line 1: A single integer that is the minimum possible length of the longest segment of Dr.Kong 's route.

Samples

input

Copy

```
7 9 2
1 2 2
2 3 5
3 7 5
```

```
1 4 1
4 3 1
4 5 7
5 7 1
1 6 3
6 7 3
```

output

Copy

```
5
```

Source

第四届河南省大学生程序设计竞赛

Submit

Codes

HZNUOJ is based on HUSTOJ

--- Maintainer List ---

Server Time: 2025-8-26 15:41:09